Empirical Evaluation of Geo-Indistinguishability Under Bayesian Linkage Attack

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Abstract

This paper evaluates the geo-indistinguishability mechanism (Andrés et al., 2013) on NYC restaurant inspection data using PLUTO as auxiliary information for a Bayesian linkage attack. We assess loss of privacy and utility and test these against theoretical bounds set by the paper. This is done using single query attacks as well as multi-record attacks that exploit linked inspections. Results show that the strongest Bayesian prior raises re-identification by 48% relative to naïve spatial proximity at $\varepsilon = 0.01$ (up to 89% at $\varepsilon = 0.002$), and that within a given setting, record-level vulnerability closely tracks noise displacement (Spearman ρ between -0.89 and -0.96), although re-identification rates still differ by borough.

1 Introduction

Privacy protection in public data releases is a long-standing challenge in data governance. Datasets containing geographic coordinates are particularly sensitive. Location information can reveal patterns of behavior, health conditions, or economic circumstances. Common anonymization strategies include aggregating records, encoding column values, or omitting identifying fields. However, when datasets contain precise coordinates, these approaches may be insufficient. Even coarsened coordinates, combined with publicly available auxiliary data, can enable re-identification of specific individuals or establishments.

Geo-indistinguishability (Andrés et al., 2013) addresses this by defining a formal privacy guarantee scaled by physical distance. This is achieved by the planar Laplace distribution, a radially symmetric 2D noise distribution. The paper proves that the guarantee is prior-independent. It bounds the attacker's information gain regardless of their auxiliary knowledge (Theorem 3.1). However, the paper's empirical evaluation is limited to mechanism-level metrics (noise displacement statistics and bandwidth overhead for LBS queries). It does not evaluate how the mechanism performs when an attacker has access to correlated auxiliary data, when repeated observations of the same location are available (the composition threat described in Section 3.3), or when the released data will be used for spatial policy analysis rather than individual queries.

This study addresses these gaps by introducing a reproducible approach to empirically evaluate location privacy under the planar Laplace mechanism. We demonstrate this approach on NYC restaurant inspection data, using the PLUTO building database as auxiliary information for a Bayesian linkage attack. A linkage attack attempts to match records in an anonymized dataset to their corresponding entries in an auxiliary dataset with known locations. Our evaluation spans three dimensions: re-identification rates under five prior configurations, utility loss for downstream spatial analyses, and posterior-to-prior belief ratios compared against the paper's theoretical bounds. We additionally evaluate a multi-record attack that exploits the CAMIS identifier linking repeated inspections, directly testing the composition guarantee from Section 3.3. While the specific results

are dataset-dependent, the methodology is portable to any geocoded data release with an available reference dataset. The core steps (identifying linkable attributes, learning priors from historical data, and measuring privacy, utility, and theoretical tightness) remain the same.

2 Background

2.1 Geo-indistinguishability

Standard differential privacy protects a row in a database by ensuring the output distribution is similar whether that row is included or not. When the "database" is a single user's location, this would require the output to be essentially independent of the true location, rendering the service useless. Geo-indistinguishability relaxes this by scaling the privacy guarantee with physical distance.

A mechanism K satisfies ε-geo-indistinguishability if for all locations x, x' and all measurable sets Z:

$$K(x)(Z) \leq e^{\varepsilon \cdot d(x,x')} \cdot K(x')(Z) \tag{1}$$

The parameter ε is in inverse distance units and controls the rate at which privacy degrades with distance. The expected displacement is $2/\varepsilon$.

The guarantee is prior-independent: for any prior π the attacker holds, the posterior-to-prior ratio is bounded by $e^{\varepsilon \cdot \text{diameter}(N)}$ for any region N (Theorem 3.1). This is the theoretical ceiling our empirical evaluation tests against.

2.2 The planar Laplace mechanism

The mechanism adds noise from a planar Laplace distribution with density proportional to $e^{-\varepsilon \cdot d}$ at distance d from the true location. Unlike independent 1D Laplace noise on each coordinate, the planar Laplace is radially symmetric, providing uniform privacy in all directions. Sampling uses polar coordinates: draw angle θ uniformly from $[0, 2\pi)$, draw radius r via inverse CDF from $C(r) = 1 - (1 + \varepsilon r)e^{-\varepsilon r}$, and add $(r \cdot \cos\theta,\ r \cdot \sin\theta)$ to the true location.

2.3 Composition for traces

Section 3.3 of the paper addresses the case where the same location generates n independent noisy observations. Under standard composition, n independent ε-geo-indistinguishable observations degrade to $n\varepsilon$ total privacy loss.

3 Data and Threat Model

3.1 Datasets

NYC DOHMH Restaurant Inspection Results. Inspection records for restaurants across all five boroughs. Each record includes a unique restaurant identifier (CAMIS), inspection date, coordinates, borough, cuisine description, violation codes, critical flag, score, and grade. Multiple inspections of the same restaurant share the same CAMIS. Coordinates were projected from WGS84 to NYC State Plane (EPSG:2263, in feet) to ensure Euclidean distances are physically meaningful.

NYC PLUTO. Tax lot records for 857,103 lots across five boroughs, including BBL identifier, coordinates (State Plane), land use code, building class, residential units, floors, year built, building area, and borough. This serves as the attacker's auxiliary information.

3.2 Threat model and data split

Inspection data was released openly through 2024 with full BBL and address information. Starting in 2025, the publisher anonymizes by adding planar Laplace noise and stripping BBL. The attacker uses historical releases (training: 2020–2024, 44,321 visits, 21,637 restaurants, of which 43,237 visits join to PLUTO by BBL) to learn priors and attacks the anonymized release (test: inspections from January 2025 onward, 37,860 visits, 24,595 restaurants). The released test data includes noisy coordinates, CAMIS, inspection date, cuisine description, borough, critical flag, score, and grade. BBL is withheld as its inclusion would make reidentification trivial.

We use PLUTO version 26v1 for both training and attack. Building stock turnover between versions is less than 0.1%, and we keep only test records whose BBL appears in PLUTO (37,860 of 38,762 geocoded test visits). BBL is used only during evaluation to verify correctness. It is never provided to the attack algorithm.

3.3 Attack methodology

The attack implements the optimal Bayesian strategy from the paper's theoretical framework. For each noisy observation z, the attacker computes

$$\text{score}(x) = \pi(x) \cdot e^{-\varepsilon \cdot d(x,z)} \tag{2}$$

for each candidate building x, where $\pi(x)$ is the prior weight and $d(x, z)$ is Euclidean distance. The highest-scoring building is the guess. Candidates are restricted to buildings within a search radius r satisfying $C(r) = 0.99$, derived from the mechanism's CDF. This is the MAP estimator for a given prior, making the evaluation a conservative upper bound on the mechanism's practical effectiveness.

3.4 Configurations

We evaluate five configurations:

- **Spatial only**: uniform prior over all PLUTO buildings. Equivalent to nearest-neighbor matching. Baseline.
- **Borough**: restrict candidates to PLUTO buildings in the matching borough. Near-binary filter.
- **Cuisine → landuse**: weight candidates by $P(\text{landuse} \mid \text{cuisine type})$. Captures that Chinese restaurants concentrate in mixed-use buildings (landuse 4.0) while hamburger chains are in standalone commercial lots (landuse 5.0).
- **Cuisine → units**: weight candidates by $P(\text{units_bin} \mid \text{cuisine type})$. Captures associations between restaurant types and building density.
- **Cuisine → year**: weight candidates by $P(\text{year_bin} \mid \text{cuisine type})$. Moderate signal.

Priors are learned by joining historical inspection records to PLUTO via BBL and computing conditional frequency tables. A smoothing value of 0.01 prevents zero probabilities. These reflect empirical associations, not causal relationships.

3.5 Multi-record attack

The multi-record attack exploits the CAMIS identifier linking multiple inspections to the same restaurant. When a restaurant is inspected n times and each inspection's coordinates are independently distorted, the attacker observes n noisy points scattered around the same true location. The attack proceeds in four steps: (1) for each noisy observation, query the PLUTO KDTree for all buildings within the search radius, (2) intersect all n candidate sets, (3) compute the joint score:

$$\log \text{score}(x) = \log \pi(x) - \varepsilon \cdot \sum_{i=1}^{n} d(x, z_i) \tag{3}$$

and (4) select the highest-scoring candidate. If the intersection is empty, the union is used as fallback.

3.6 General approach

While our evaluation uses NYC-specific datasets, the methodology generalizes to any geocoded data release. The steps are: (1) identify a reference dataset with precise locations and building attributes (the auxiliary data), (2) identify attributes in the released dataset that correlate with reference dataset attributes, (3) learn conditional frequency tables from historical releases where the join key (e.g., BBL) is available, (4) apply the planar Laplace mechanism at varying ε, (5) run the Bayesian linkage attack under each prior configuration, and (6) evaluate re-identification rates, utility distortion, and posterior-to-prior ratios. Steps 2–3 are dataset-specific; the remaining steps are mechanistic. The choice of utility metrics (step 6) should reflect the intended downstream use of the data.

4 Evaluation Metrics

Re-identification rate. Fraction of test records for which the attacker correctly identifies the true PLUTO building (verified by BBL match).

Utility loss. Two forms of distortion. Inspection score distortion: RMSE and MAE of mean inspection scores per census tract, true vs. noisy coordinates. Cuisine composition distortion: total variation distance (TVD) between per-tract cuisine distributions, true vs. noisy. TVD ranges from 0 (identical) to 1 (completely different).

Posterior-to-prior ratio. $P(\text{true building} \mid z)$ / $\pi(\text{true building})$ for each record, measuring how much the attacker's belief shifted after observing the noisy coordinates. Compared against the theoretical bound $e^{\varepsilon \cdot d}$ from Theorem 3.1.

5 Results

We test ε values ranging from 0.001 to 0.05 ft^{-1}. Values above 0.003 fall outside the range explored by Andrés et al. (corresponding to $\ell = \ln(2)$ to $\ln(6)$ at $r = 200\text{m}$). Higher values are included to characterize the full privacy-utility tradeoff.

5.1 Single-query re-identification

Re-identification rates rise monotonically with ε. Cuisine type paired with land use is the strongest prior, and even it provides only modest absolute improvement. The units and year-built priors

perform at or below the spatial-only baseline at most ε (for example, 8.37% and 8.09% vs 9.03% at $\varepsilon = 0.01$). At $\varepsilon = 0.01$, cuisine $\rightarrow$ landuse achieves 13.37% re-identification compared to the 9.03% spatial-only baseline, a 48.0% relative improvement. At $\varepsilon = 0.05$, the gap narrows to 15.0% relative lift (29.50% vs 25.65%). Borough provides virtually no improvement over spatial-only, confirming that geographic filtering is redundant with spatial proximity when the search radius is smaller than a borough.

The relative lift from the cuisine $\rightarrow$ landuse prior peaks at 89.2% ($\varepsilon = 0.002$) and declines steadily as ε increases. It is lower at the noisiest setting (70.7% at $\varepsilon = 0.001$), but across most of the range, priors are most valuable when the spatial signal is weak. When noise is small and the spatial likelihood is already sharp, the prior adds marginal value.

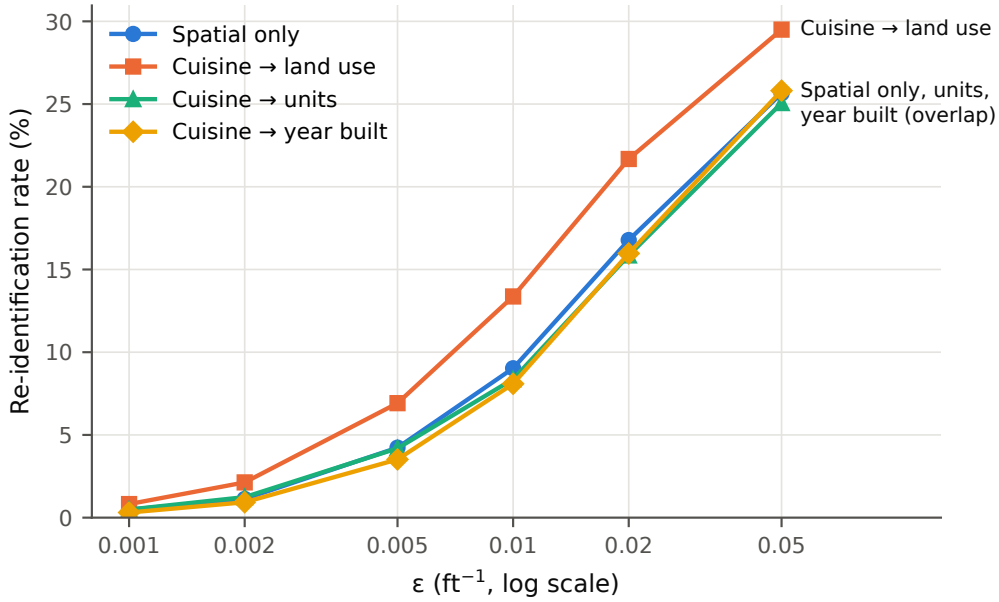


Figure 1: Single-query re-identification rate vs ε by prior configuration (n = 37,860 test visits). Borough-only is omitted because it is within 0.02 percentage points of spatial-only at every ε.

5.2 Posterior analysis

The theoretical bound from Theorem 3.1 is satisfied in 100% of tested records, and it is far from tight. Median empirical posterior-to-prior ratios range from 1.1 to 3.8 across configurations and ε (2.4 to 3.8 for $\varepsilon \leq 0.01$), and the largest ratio observed for any record is 290 (cuisine $\rightarrow$ year, $\varepsilon = 0.001$). For comparison, the search radius r satisfies $\varepsilon r \approx 6.64$ at every ε, so a record's bound $e^{\varepsilon D}$, where D is the distance from the true building to the farthest candidate, is about $e^{6.64} \approx 760$ or more whenever $D \geq r$, which holds when candidates fill the search disc. The bound therefore sits more than two orders of magnitude above typical empirical ratios. Median attacker confidence at the true building $P(\text{true} \mid z)$ is 0.03% at $\varepsilon = 0.001$ and about 2% at $\varepsilon = 0.01$ (spatial only). At $\varepsilon \leq 0.005$, fewer than 0.4% of records have $P(\text{true} \mid z) > 0.5$ under any configuration. This rises to 0.5–2.0% at $\varepsilon = 0.01$ and to 20–28% at $\varepsilon = 0.05$, the weakest privacy setting tested. At $\varepsilon \leq 0.01$, then, the attacker's confidence in the true building stays low in absolute terms, even though median posterior-to-prior ratios of 2.4 to 3.8 mean that confidence roughly triples.

Within a given setting, the strongest predictor of record-level vulnerability is the magnitude of the noise draw. Across the 20 combinations of attack configuration and $\varepsilon \leq 0.01$, the Spearman

correlation between noise displacement and posterior-to-prior ratio ranges from -0.89 to -0.96 ($p < 0.001$ in every case). Much of the record-to-record variation in risk is therefore stochastic, driven by a small fraction of records receiving little noise by chance. These are the records an attacker can re-identify regardless of context. Risk is not uniform across places, however: re-identification rates differ by borough under the same ε (Section 6).

5.3 Multi-record attack

Releasing a linkable identifier (CAMIS) measurably degrades privacy, but the degradation is less severe than the theoretical $n \cdot \varepsilon$ bound predicts. For spatial-only at $\varepsilon = 0.01$: $n = 1$ gives 9.0%, $n = 2$ gives 11.2%, $n = 3$ gives 13.9%, $n = 4$ gives 14.2%. With cuisine $\rightarrow$ landuse: $n = 1$ gives 13.4%, $n = 3$ gives 18.7%. The gains plateau around $n = 3$–4, although few restaurants have more than four inspections, so estimates beyond that are imprecise (Figure 2).

To test the composition bound, we compare multi-record re-identification to single-query at equivalent effective ε. For spatial-only: $n = 2$ at $\varepsilon = 0.005$ (effective $\varepsilon = 0.01$) achieves 4.9%, compared to 9.0% for a single query at $\varepsilon = 0.01$. The multi-record attack achieves substantially less than the theoretical worst case. The intersection constraint, which requires candidates to be spatially consistent across all observations, provides additional protection the bound does not account for.

The practical policy implication: at $\varepsilon = 0.01$, releasing CAMIS costs up to about 5 percentage points of additional re-identification risk under the strongest prior: 13.4% for single observations, 15.9% for restaurants with two linked inspections (the most common case), and 18.7% for restaurants with three. A data publisher can weigh this cost against the analytical benefit of tracking inspection histories.

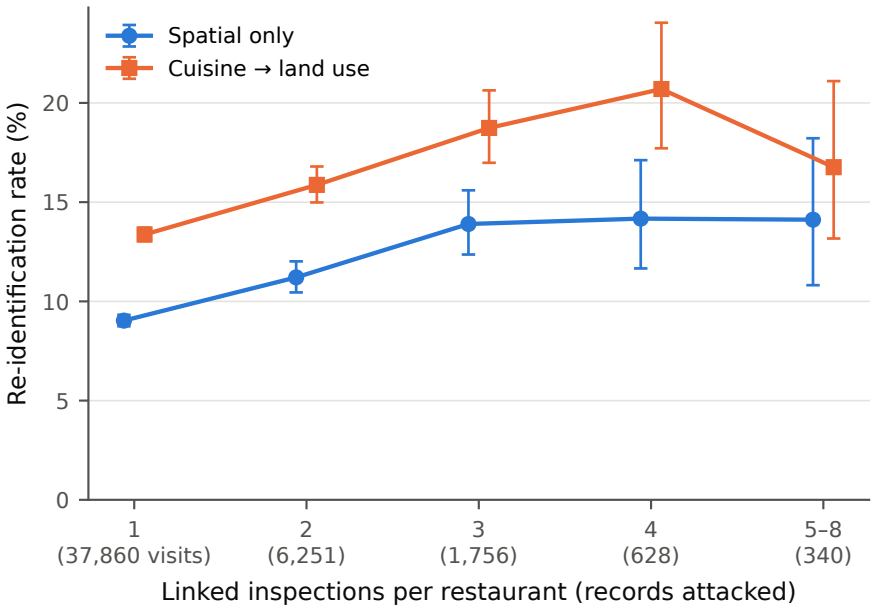


Figure 2: Multi-record attack at $\varepsilon = 0.01$: re-identification rate by number of linked inspections. $n = 1$ is the single-query attack on all test visits; $n \geq 2$ covers restaurants with exactly n inspections (counts in parentheses), with $n = 5$–8 pooled (the 8 restaurants with more than eight inspections are excluded). Error bars are 95% Wilson intervals across records for a single noise draw.

5.4 Utility loss

The mechanism introduces substantial analytical distortion at privacy levels where re-identification risk remains moderate. At $\varepsilon = 0.01$, mean cuisine composition TVD is 0.34 (median 0.30; roughly a third of each tract's cuisine distribution changes) and inspection score RMSE is 8.09 points. At $\varepsilon = 0.001$, mean TVD rises to 0.59 and RMSE to 11.11. Data publishers face a quantifiable tradeoff: at $\varepsilon = 0.01$, 87% of test records are not re-identified by the strongest attack, while mean cuisine distortion is 0.34 TVD per tract.

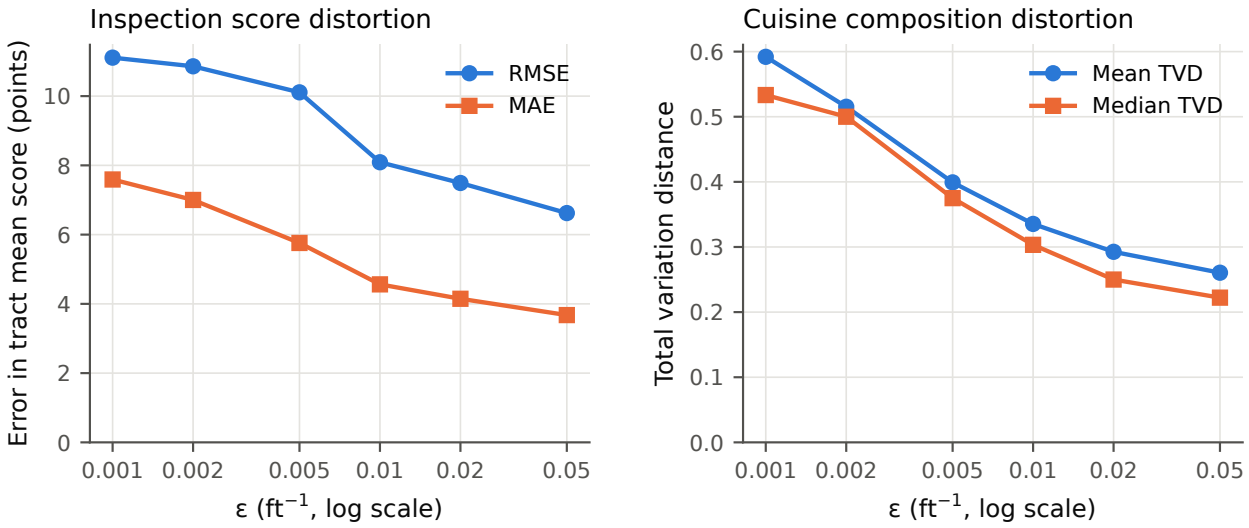


Figure 3: Utility loss vs ε, computed over 2020 Census tracts. Left: error in tract mean inspection score (RMSE and MAE). Right: total variation distance between true and noisy tract cuisine distributions (mean and median across tracts).

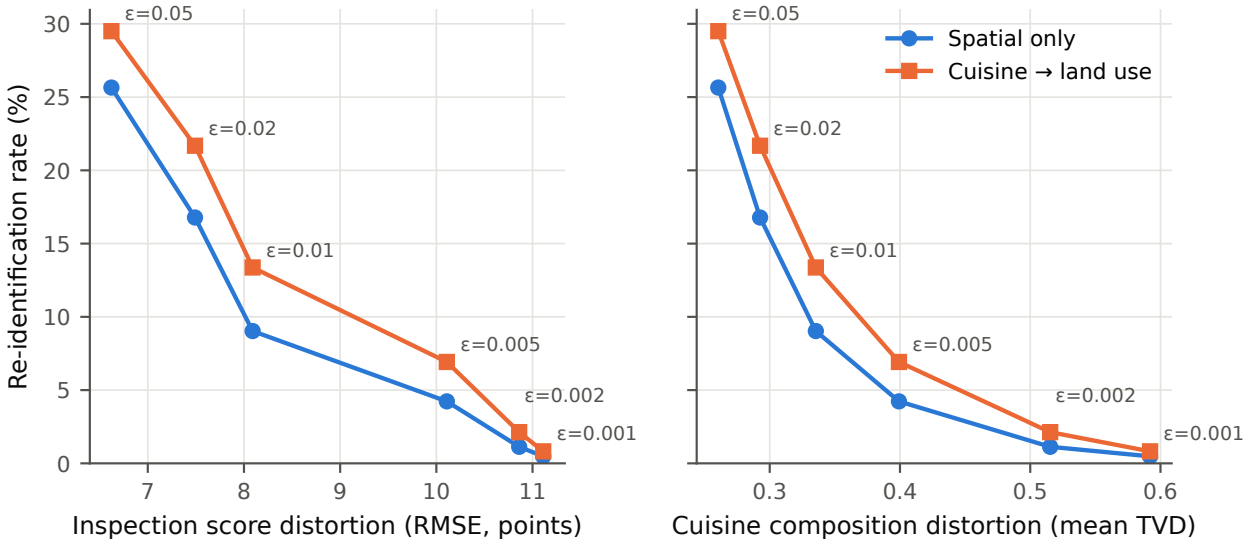


Figure 4: Privacy-utility tradeoff. Re-identification rate against utility distortion for the spatial-only and cuisine → landuse attacks; each point is one ε (labeled on the cuisine → landuse series).

6 Discussion

The key insight is that the mechanism's effectiveness depends less on the attacker's auxiliary information and more on the stochastic noise draw, which the mechanism's design already accounts for. Each of our three metrics supports this from a different angle.

Re-identification rates show that even the strongest prior (cuisine → landuse) yields at most 4.9 percentage points of absolute improvement over spatial-only matching. The relative value of priors diminishes as ε increases, from 89% lift at $\varepsilon = 0.002$ to 15% at $\varepsilon = 0.05$. This indicates that spatial proximity, not auxiliary knowledge, is the primary driver of attack success.

The posterior analysis confirms that the theoretical bound from Theorem 3.1 holds for every tested record and sits more than two orders of magnitude above typical empirical ratios. The strong correlation between noise displacement and posterior-to-prior ratio (Spearman ρ of -0.89 to -0.96)

The multi-record attack shows that the composition bound $n \cdot \varepsilon$ similarly overestimates empirical degradation. The spatial intersection constraint provides protection the worst-case theory does not capture, and gains plateau after 3–4 observations.

At the same time, utility loss is substantial. Even at $\varepsilon = 0.01$, cuisine composition distortion (mean TVD = 0.34) would meaningfully affect tract-level food access analyses. Publishers should select ε based on the specific downstream analyses they need to preserve, using the privacy-utility curves this methodology produces.

The utility metric is the most customizable and will heavily influence the optimal epsilon value for a given dataset.

Limitations. This evaluation is specific to NYC's dense urban environment and may not generalize to suburban or rural settings where candidate density is lower. The smoothing parameter (0.01) in prior construction affects the balance between prior and spatial likelihood. The independence assumption when combining priors is violated in practice, which is why we test each individually. We evaluate a single mechanism (planar Laplace) and do not compare against alternatives such as deterministic rounding or the exponential mechanism. We also limited candidate

sets to reduce computational costs (restricted radius and intersection requirement in multi-query attacks), so removing these restrictions could yield higher attack success.

Future work. Natural extensions include spatially adaptive ε calibrated to local candidate density, posterior confidence ($P(\text{true} \mid z) > \text{threshold}$) as a stricter privacy metric, extension to non-Euclidean metric spaces (d_X-privacy, Chatzikokolakis et al., 2013), and a one-to-one bipartite matching attack enforcing the constraint that each building can only be matched once.

7 References

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